



GATE 2027 · PREMIUM FORMULA SHEET

Engineering Mathematics Formula Sheet

Complete Coverage — All 6 Branches

Covers Linear Algebra, Calculus, Differential Equations, Complex Analysis, Probability & Statistics, Vector Analysis, Discrete Mathematics, and Numerical Methods — mapped to every GATE 2027 branch syllabus. Every formula: *What · Why · When to use.*

CS

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Matrix Operations & Properties

Concept	Formula / Property	Why / Meaning	When to Use
Matrix Addition	$A+B: (A+B)_{ij} = A_{ij}+B_{ij}$	Element-wise; only same-size matrices can add.	Superposition, combining linear maps.
Matrix Multiplication	$(AB)_{ij} = \sum_k A_{ik}B_{kj}$	Row×Column dot products; $A(m \times n) \times B(n \times p) = (m \times p)$.	Composing linear transformations, systems.
Transpose	$(A^T)_{ij} = A_{ji}; (AB)^T = B^T A^T$	Rows become columns; reverses multiplication order.	Symmetric test: $A=A^T$; orthogonality; quadratic forms.
Trace	$\text{tr}(A) = \sum a_{ii} = \sum \lambda_i$	Sum of diagonal = sum of eigenvalues.	Quick eigenvalue check; invariant under similarity.
Determinant (2×2)	$ A = ad-bc$	Signed area scaling of the linear map.	Invertibility check: $ A \neq 0$ iff A is invertible.
Determinant (3×3) Cofactor	$ A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$	Expand along any row/column; $C_{ij} = (-1)^{(i+j)}M_{ij}$.	3×3 determinant calculation.
Properties of Determinant	$ AB = A B ; A^T = A ; kA = k^n A $	Multiplicativity; transpose invariance; scalar scaling.	Simplifying determinant computations.
Inverse (2×2)	$A^{-1} = (1/ A) \cdot [d \ -b; -c \ a]$	Adjugate divided by determinant.	Solving 2×2 systems; direct inverse.
Inverse (General)	$A^{-1} = \text{adj}(A)/ A ; A \cdot A^{-1} = I$	Exists iff $ A \neq 0$ (non-singular).	Solving $AX=B$: $X=A^{-1}B$.



Eigenvalues & Eigenvectors

Concept	Formula	Why / Meaning	When to Use
Characteristic Equation	$\det(A - \lambda I) = 0$	Eigenvalues λ are scalings where $Av = \lambda v$ ($v \neq 0$).	Finding all eigenvalues of a matrix.

Concept	Formula	Why / Meaning	When to Use
Eigenvector	$(A - \lambda I)v = 0 \rightarrow \text{solve null space}$	Non-trivial solution exists when $\det=0$.	Find eigenvector for each eigenvalue λ .
Trace = $\sum \lambda_i$	$\text{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$	Quick verification after finding eigenvalues.	Checking eigenvalue calculation.
$\det(A) = \prod \lambda_i$	$ A = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n$	Product of all eigenvalues equals determinant.	Verify eigenvalues; check singularity.
Cayley-Hamilton	A satisfies its own char. poly: $p(A)=0$	Every matrix satisfies its characteristic polynomial.	Computing powers of A, A^{-1} via polynomial.
Diagonalization	$A = PDP^{-1}$; $D = \text{diag}(\lambda_1, \dots, \lambda_n)$	P=eigenvector matrix; valid if n linearly independent eigenvectors.	Computing A^n , solving systems of ODEs.
LU Decomposition	$A = LU$; L=lower triangular, U=upper triangular	Factors matrix for efficient Gaussian elimination.	Solving $AX=b$ multiple times with same A.



Rank, Nullity & Linear Systems

Concept	Formula / Rule	Why / Meaning	When to Use
Rank-Nullity Theorem	$\text{rank}(A) + \text{nullity}(A) = n$ (cols)	Dimensions of column space and null space sum to total columns.	Finding null space dimension given rank.
Solution of $AX = b$	Unique: $\text{rank}(A) = \text{rank}([A b]) = n$ Infinite: $\text{rank}(A) = \text{rank}([A b]) < n$ None: $\text{rank}(A) \neq \text{rank}([A b])$	Rouché-Capelli theorem for consistency.	Classifying solutions of any linear system.
Homogeneous $AX=0$	Always has trivial sol $X=0$; non-trivial iff $\text{rank}(A) < n$	Non-trivial solutions exist only if $\det(A)=0$.	Eigenvalue problems; testing linear dependence.



ALL BRANCHES



Limits, Continuity & Differentiability

Concept	Formula / Theorem	Why / Meaning	When to Use
L'Hôpital's Rule	$\lim f/g \text{ [} 0/0 \text{ or } \infty/\infty \text{]} = \lim f'/g'$	Resolves indeterminate forms by differentiating numerator and denominator.	$0/0$, ∞/∞ limits; apply repeatedly if still indeterminate.
Standard Limits	$\lim_{x \rightarrow 0} \sin x/x = 1$; $(1+1/n)^n \rightarrow e$; $(e^x - 1)/x \rightarrow 1$	Fundamental limit results used constantly.	Evaluating trig and exponential limits.
Continuity at $x=a$	$f(a)$ exists ; $\lim f(x)=f(a)$; LHL=RHL	No breaks, jumps, or holes in graph.	Checking if piecewise function is continuous.
Mean Value Theorem	$\exists c \in (a,b) : f'(c) = (f(b)-f(a))/(b-a)$	Instantaneous rate equals average rate at some point.	Proving existence results; root isolation.
Rolle's Theorem	$f(a)=f(b) \rightarrow \exists c \in (a,b) : f'(c)=0$	Special case of MVT: horizontal tangent exists between equal endpoints.	Proving at least one critical point between equal values.



Derivatives — Key Formulas

Rule / Function	Derivative	When to Use
Power Rule	$d/dx(x^n) = nx^{n-1}$	Any polynomial term.
Product Rule	$d/dx(uv) = u'v + uv'$	Product of two functions.
Quotient Rule	$d/dx(u/v) = (u'v - uv')/v^2$	Fraction of two functions.
Chain Rule	$d/dx[f(g(x))] = f'(g(x)) \cdot g'(x)$	Composite functions; essential for all implicit problems.
Exponential / Log	$d/dx(e^x)=e^x$; $d/dx(\ln x)=1/x$; $d/dx(a^x)=a^x \ln a$	Exponential growth/decay models.
Trig Functions	$\sin \rightarrow \cos$; $\cos \rightarrow -\sin$; $\tan \rightarrow \sec^2$; $\cot \rightarrow -\csc^2$	Oscillatory systems, signal analysis.

Rule / Function	Derivative	When to Use
Inverse Trig	$\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}} ;$ $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{(1+x^2)}$	Integration of algebraic expressions via inverse trig substitution.

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Integration

Technique / Formula	Rule	When to Use
Power Rule (integral)	$\int x^n dx = \frac{x^{n+1}}{(n+1)} + C \quad [n \neq -1]$	Polynomial integration.
$\int \frac{1}{x} dx$	$\ln x + C$	Rational functions with linear denominator.
$\int e^x dx ; \int a^x dx$	$e^x + C ; \frac{a^x}{\ln(a)} + C$	Exponential functions.
Integration by Parts	$\int u dv = uv - \int v du$ (LIATE rule for u)	Products: Logs, Inverse trig, Algebraic, Trig, Exponential.
Substitution	Let $u=g(x)$; $\int f(g(x))g'(x)dx = \int f(u)du$	Composite integrands; chain rule in reverse.
Definite Integral (FTC)	$\int_a^b f(x)dx = F(b)-F(a)$ where $F'=f$	Area under curve, accumulated change.
Improper Integral	$\int_1^\infty (1/x^p)dx$ converges iff $p>1$	Infinite limits or discontinuous integrands.
Taylor Series	$f(x) = \sum \frac{f^{(n)}(a)}{n!} \cdot (x-a)^n$	Approximating functions; series solutions of ODEs.
Maclaurin Series (key)	$e^x = \sum x^n/n!$; $\sin x = x - x^3/6 + \dots$; $\cos x = 1 - x^2/2 + \dots$	Limit evaluation; approximate solutions.
Maxima & Minima	$f'(x)=0 \rightarrow$ critical pt; $f''>0 \rightarrow$ min; $f''<0 \rightarrow$ max; $f''=0 \rightarrow$ test higher derivatives	Optimization problems in any branch.



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Partial Derivatives

Concept	Formula	Why / Meaning	When to Use
Partial Derivative	$\partial f / \partial x$: differentiate f w.r.t. x , treat y as constant	Rate of change in one direction of a multivariable function.	Multivariable optimization, PDEs.
Total Derivative	$df = (\partial f / \partial x)dx + (\partial f / \partial y)dy$	Total change combining all partial changes.	Error analysis, sensitivity, chain rule in 2D.
Hessian Matrix	$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$	Second-order info for 2D optimization.	Classify critical points in 2 variables.
Saddle / Max / Min (2D)	$D = f_{xx} \cdot f_{yy} - (f_{xy})^2$; $D > 0, f_{xx} > 0 \rightarrow \text{min}$; $D > 0, f_{xx} < 0 \rightarrow \text{max}$; $D < 0 \rightarrow \text{saddle}$	Discriminant determines critical point type.	2-variable optimization (ME, CE structures).
Jacobian	$J = \partial(u, v) / \partial(x, y) = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$	Scaling factor for variable change in double integrals.	Coordinate transformations: Cartesian $\rightarrow$ Polar etc.



Vector Calculus

Operator / Theorem	Formula	Why / Meaning	When to Use
Gradient (∇f)	$\nabla f = (\partial f / \partial x)\hat{i} + (\partial f / \partial y)\hat{j} + (\partial f / \partial z)\hat{k}$	Points in direction of steepest ascent; magnitude = max rate of change.	Directional derivative, potential fields.
Directional Derivative	$D_{\hat{u}} f = \nabla f \cdot \hat{u}$	Rate of change of f in direction of unit vector $\hat{u}$.	Rate of temperature/potential change in given direction.
Divergence ($\nabla \cdot F$)	$\text{div } F = \partial P / \partial x + \partial Q / \partial y + \partial R / \partial z$	Net outward flux per unit volume; source/sink measure.	Gauss's theorem, fluid flow, EM fields.
Curl ($\nabla \times F$)	$\text{curl } F = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial / \partial x & \partial / \partial y & \partial / \partial z \\ P & Q & R \end{vmatrix}$	Rotation/circulation of vector field.	Stokes' theorem, magnetic fields, rotation analysis.



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First Order ODEs

Type	Standard Form / Solution Method	Why / Key Step	When to Use
Separable ODE	$dy/dx = f(x)g(y) \rightarrow \int dy/g(y) = \int f(x)dx$	Separate x and y terms; integrate both sides.	Growth/decay, Newton's cooling.
Linear First-Order ODE	$dy/dx + P(x)y = Q(x)$; IF = $e^{\int P dx}$; $y \cdot IF = \int Q \cdot IF dx$	Integrating Factor converts LHS to exact derivative $d/dx(y \cdot IF)$.	RC circuits, first-order systems.
Exact ODE	$Mdx + Ndy = 0$; exact if $\partial M/\partial y = \partial N/\partial x$; F s.t. $\partial F/\partial x = M$, $\partial F/\partial y = N$	Total differential of some $F(x,y)$; solution $F(x,y) = C$.	Conservative field equations.
Bernoulli Equation	$dy/dx + P(x)y = Q(x)y^n$; let $v = y^{1-n} \rightarrow$ linear in v	Substitution linearizes the nonlinear term.	Population models with inhibition.



Higher Order Linear ODEs (Constant Coefficients)

Concept	Formula / Method	Why / Meaning	When to Use
General Solution	$y = y_h + y_p$ (homogeneous + particular)	Linear ODE solution is the complementary function plus any particular solution.	All higher-order linear ODEs.
Characteristic Equation	For $a_n y^{(n)} + \dots + a_0 y = 0$; substitute $y = e^{rx} \rightarrow a_n r^n + \dots + a_0 = 0$	Converts ODE to algebraic equation in r.	Finding complementary function y_h .
Real distinct roots ($r_1 \neq r_2$)	$y_h = C_1 e^{r_1 x} + C_2 e^{r_2 x}$	Two independent exponential solutions.	Overdamped systems.
Repeated real roots ($r_1 = r_2 = r$)	$y_h = (C_1 + C_2 x) e^{rx}$	Multiply by x to get second independent solution.	Critically damped systems.
Complex roots ($\alpha \pm \beta i$)	$y_h = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$	Euler's formula $e^{i\beta x} = \cos \beta x + i \sin \beta x$.	Underdamped oscillatory systems, RLC circuits.

Concept	Formula / Method	Why / Meaning	When to Use
Variation of Parameters	$y_p = -y_1 \int (y_2 g/W) dx + y_2 \int (y_1 g/W) dx;$ $W = \text{Wronskian}$	General method for any $g(x)$ on RHS; Wronskian $W = y_1 y_2' - y_2 y_1'$.	When RHS is not a polynomial, trig, or exponential.
Cauchy-Euler Equation	$x^2 y'' + ax y' + by = 0;$ let $y = x^m \rightarrow m^2 + (a-1)m + b = 0$	Substitution $x = e^t$ or $y = x^m$ converts to constant-coefficient ODE.	Equations with variable coefficients $x^n \cdot y^{(n)}$ form.



Laplace Transforms (ME, EE)

$f(t)$	$L\{f(t)\} = F(s)$	When to Use
1 (unit step)	$1/s$	Solving IVPs with discontinuous/impulsive forcing; converting ODE to algebraic equation in s-domain.
t	$1/s^2$	
t^n	$n!/s^{(n+1)}$	
e^{at}	$1/(s-a)$	
$\sin(at)$	$a/(s^2+a^2)$	
$\cos(at)$	$s/(s^2+a^2)$	
Derivative $L\{f'\}$	$sF(s) - f(0)$	



Partial Differential Equations (PDE)

Equation	Form	Method of Solution	Branch
Heat Equation (1D)	$\partial u / \partial t = \alpha^2 \cdot \partial^2 u / \partial x^2$	Separation of variables $\rightarrow X(x)T(t)$; Fourier series for BCs.	CE, ME, EC
Wave Equation (1D)	$\partial^2 u / \partial t^2 = c^2 \cdot \partial^2 u / \partial x^2$	Separation of variables; d'Alembert's solution.	CE, ME, EC
Laplace Equation (2D)	$\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0$	Separation of variables; complex potentials.	EE, CE, EC
Fourier Series	$f(x) = a_0/2 + \sum [a_n \cos(n\pi x/L) + b_n \sin(n\pi x/L)];$ $a_n = (1/L) \int f \cos dx;$ $b_n = (1/L) \int f \sin dx$	Representing periodic functions; BCs in PDE.	CE, EE, ME





EC

EE

ME



Complex Numbers & Analytic Functions

Concept	Formula	Why / Meaning	When to Use
Complex Number	$z = x+iy$; $ z =\sqrt{(x^2+y^2)}$; $\arg(z)=\tan^{-1}(y/x)$	Polar form: $z=r\cdot e^{i\theta}$ where $r= z $, $\theta=\arg(z)$.	All complex variable problems.
Euler's Formula	$e^{i\theta} = \cos\theta + i \sin\theta$	Bridges exponential and trigonometric functions.	Converting between polar and rectangular; phasors in EE.
De Moivre's Theorem	$(\cos\theta+i \sin\theta)^n = \cos(n\theta)+i \sin(n\theta)$	n th power of unit complex number rotates angle by n .	Finding n th roots; powers of complex numbers.
Cauchy-Riemann Equations	$\frac{\partial u}{\partial x}=\frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y}=-\frac{\partial v}{\partial x}$	Necessary condition for $f(z)=u+iv$ to be analytic (holomorphic).	Testing analyticity of complex functions.
Analytic Function	f is analytic at z_0 if C-R eqs hold and partials are continuous	Complex differentiable; satisfies Laplace equation (u and v are harmonic).	Conformal mapping, complex integration.
Harmonic Function	$\nabla^2 u = u_{xx} + u_{yy} = 0$	Real/imaginary part of analytic function satisfies Laplace equation.	Potential theory, steady-state heat.

🎵 Complex Integration & Residues

Theorem	Formula / Statement	Why / Meaning	When to Use
Cauchy's Integral Theorem	$\oint_C f(z)dz = 0$ (f analytic inside/on C)	Analytic functions have zero circulation — path independence.	Showing integral vanishes; deforming contours.
Cauchy's Integral Formula	$f(a) = (1/2\pi i) \oint_C f(z)/(z-a) dz$	Value of analytic function determined by boundary values.	Evaluating contour integrals with simple poles.

Theorem	Formula / Statement	Why / Meaning	When to Use
nth Derivative Formula	$f^{(n)}(a) = n!/(2\pi i) \oint_C \frac{f(z)}{(z-a)^{n+1}} dz$	Generalization of Cauchy's formula to higher derivatives.	Computing derivatives via contour integration.
Laurent Series	$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-z_0)^n \quad (n \text{ from } -\infty \text{ to } \infty)$	Generalized Taylor series valid in annular region; captures poles via negative powers.	Expanding functions with isolated singularities.
Residue at Simple Pole	$\text{Res}[f, z_0] = \lim_{z \rightarrow z_0} \{ (z-z_0) f(z) \}$	Coefficient of $(z-z_0)^{-1}$ in Laurent expansion.	Evaluating real integrals via residue theorem.
Residue Theorem	$\oint_C f(z) dz = 2\pi i \cdot \sum \text{Res}[f, z_k] \quad (\text{poles inside } C)$	Sum of residues at enclosed poles gives total integral.	Evaluating definite integrals of form $\int_{-\infty}^{\infty}$ or $\int_0^{2\pi}$.
Types of Singularities	Removable: $\lim f(z)$ exists; Pole of order m : $1/(z-z_0)^m$ term; Essential: infinite principal part	Classification determines residue calculation method.	Before applying residue theorem.

📌 **Series Convergence Tests:** Ratio Test: $\lim |a_{n+1}/a_n| < 1 \rightarrow$ converges. Root Test: $\lim |a_n|^{1/n} < 1 \rightarrow$ converges. Comparison Test: if $|a_n| \leq b_n$ and $\sum b_n$ converges, so does $\sum a_n$. Alternating Series: converges if $|a_n| \downarrow 0$.





ALL BRANCHES



Core Probability & Distributions

Distribution	PMF / PDF	Mean / Variance	When to Use
Bernoulli(p)	$P(X=1)=p; P(X=0)=1-p$	$\mu=p; \sigma^2=p(1-p)$	Single binary trial.
Binomial B(n,p)	${}^nC_k p^k (1-p)^{n-k}$	$\mu=np; \sigma^2=np(1-p)$	n independent binary trials.
Poisson Pois(λ)	$e^{-\lambda} \lambda^k / k!$	$\mu=\lambda; \sigma^2=\lambda$	Rare events; large n, small p; $\lambda=np$.
Uniform U(a,b)	$f(x)=1/(b-a); a \leq x \leq b$	$\mu=(a+b)/2; \sigma^2=(b-a)^2/12$	Equally likely continuous outcomes.
Exponential Exp(λ)	$f(x)=\lambda e^{-\lambda x}; x \geq 0$	$\mu=1/\lambda; \sigma^2=1/\lambda^2$	Time between Poisson events; memoryless property.
Normal N(μ, σ^2)	$f(x)=(1/\sigma\sqrt{2\pi})e^{-(x-\mu)^2/2\sigma^2}$	μ, σ^2	Natural phenomena; CLT approximation.
Standard Normal Z	$Z=(X-\mu)/\sigma; N(0,1)$	$\mu=0; \sigma^2=1$	Standardize normal for tables; z-test.
t-distribution	$t=\bar{X}/(s/\sqrt{n})$ with $v=n-1$ df	Heavier tails than normal; for small samples.	Small-sample mean testing when σ unknown.
Chi-Squared $\chi^2(k)$	Sum of k squared standard normals	$\mu=k; \sigma^2=2k$	Variance testing; goodness-of-fit.



Statistical Measures & Tests

Measure / Test	Formula	Why / Meaning	When to Use
Sample Mean	$\bar{X} = \sum x_i / n$	Unbiased estimator of population mean.	All descriptive statistics.
Sample Variance	$s^2 = \sum (x_i - \bar{X})^2 / (n-1)$	Bessel's correction for unbiased estimate.	Standard deviation calculation from sample.
Correlation Coefficient	$r = \text{Cov}(X,Y) / (\sigma_x \cdot \sigma_y);$ range $[-1,1]$	Measures linear relationship strength; ± 1 =perfect, 0=none.	Regression, signal correlation (EE, DA).

Measure / Test	Formula	Why / Meaning	When to Use
Covariance	$\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$	Joint variability; $\text{Cov}(X, X) = \text{Var}(X)$.	Multivariate statistics; DA branch.
Linear Regression	$\hat{y} = a + bx;$ $b = \text{Cov}(X, Y) / \text{Var}(X);$ $a = \bar{Y} - b\bar{X}$	Best fit line minimizing squared residuals (OLS).	Prediction, CE settlement, DA models.
Central Limit Theorem	$\bar{X} \sim N(\mu, \sigma^2/n) \text{ for large } n$	Sample means are approximately normal regardless of population distribution.	Justifying use of z-test for large samples.
Confidence Interval (z)	$\bar{X} \pm z_{\{\alpha/2\}} \cdot \sigma/\sqrt{n}$	Range capturing true mean with $(1-\alpha) \times 100\%$ probability.	Large-sample interval estimation (DA, CE).
Bayes' Theorem	$P(A B) = P(B A)P(A)/P(B)$	Updates prior probability with evidence B.	CS, DA: classification, spam detection, diagnosis.
Conditional Expectation	$E[X Y=y] = \sum x \cdot P(X=x Y=y)$	Expected value given partial information.	DA: Bayesian networks, hidden Markov models.





CS

DA

└ Propositional & First Order Logic

Concept	Symbol / Formula	Meaning	When to Use
Negation	$\neg P$ (NOT P)	Flips truth value.	Contrapositive, De Morgan's laws.
Conjunction	$P \wedge Q$ (AND)	True only when both P and Q are true.	Both conditions must hold.
Disjunction	$P \vee Q$ (OR)	True when at least one is true.	Either condition sufficient.
Implication	$P \rightarrow Q$; false only when $P=T, Q=F$	If P then Q ; contrapositive: $\neg Q \rightarrow \neg P$.	All logical deduction; equivalences.
Biconditional	$P \leftrightarrow Q$; true when P and Q have same value	P if and only if Q ; iff.	Necessary and sufficient conditions.
De Morgan's Laws	$\neg(P \wedge Q) = \neg P \vee \neg Q$; $\neg(P \vee Q) = \neg P \wedge \neg Q$	Negate conjunction/disjunction by distributing negation.	Simplifying logical expressions, gate circuits.
Tautology	True for ALL truth assignments	Universally valid formula (e.g., $P \vee \neg P$).	Identifying valid logical arguments.
Universal Quantifier	$\forall x P(x)$: $P(x)$ is true for all x in domain	First-order logic over a domain.	All-statements; proving for every element.
Existential Quantifier	$\exists x P(x)$: $P(x)$ is true for at least one x	There-exists statement.	Finding one example/counterexample.



Sets, Relations & Functions

Concept	Formula / Definition	When to Use
Set Operations	$A \cup B, A \cap B, A \setminus B, A^c, A \Delta B = (A \setminus B) \cup (B \setminus A)$	Counting, inclusion-exclusion, database queries.
Power Set	$P(A)$: all subsets; $ P(A) = 2^{ A }$	Counting subsets; bit-vector representation.

Concept	Formula / Definition	When to Use
Cartesian Product	$A \times B = \{(a,b) : a \in A, b \in B\};$ $ A \times B = A B $	Relations defined as subsets of $A \times B$.
Equivalence Relation	Reflexive + Symmetric + Transitive	Partitioning sets; congruence classes.
Partial Order	Reflexive + Antisymmetric + Transitive	Hasse diagrams; lattices; scheduling.
Injection (one-to-one)	$f(a)=f(b) \rightarrow a=b$	No two inputs map to same output.
Surjection (onto)	$\forall y \in B, \exists x : f(x)=y$	Every output is achieved.
Bijection	Both injective and surjective $\rightarrow$ invertible	Counting; inverse functions.



Graph Theory

Concept	Formula / Property	When to Use
Handshaking Lemma	$\sum \deg(v) = 2 E $	Sum of degrees = twice number of edges. Check validity.
Euler Path/Circuit	Euler circuit: all vertices even degree; Path: exactly 2 odd degree vertices	Traversal problems, routing.
Hamiltonian Path/Cycle	Visits every vertex exactly once; no simple formula (NP-complete to decide)	TSP, scheduling problems.
Tree Properties	n vertices, $n-1$ edges; connected; no cycles	Spanning trees; hierarchical data.
Chromatic Number $\chi(G)$	Min colors to color graph with no adjacent vertices same color	Scheduling, register allocation.
Bipartite Graph	G is bipartite $\leftrightarrow G$ has no odd-length cycles $\leftrightarrow 2$ -colorable	Matching problems, network flows.





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⊕ Groups & Algebraic Structures (CS/DA)

Structure	Axioms	Example	When to Use
Group $(G, \cdot)$	Closure, Associativity, Identity, Inverse	$(\mathbb{Z}, +)$; $(\mathbb{Z}_n, \times)$ if $\gcd=1$	Symmetry, cryptography, automata.
Abelian Group	Group + Commutativity ($a \cdot b = b \cdot a$)	$(\mathbb{Z}, +)$, $(\mathbb{R}, +)$	Most number-based groups in GATE.
Monoid	Closure + Associativity + Identity (no inverse needed)	$(\mathbb{N}, \times)$	String concatenation, formal languages.
Lagrange's Theorem	$ H $ divides $ G $ (H subgroup of finite G)	—	Finding possible subgroup sizes.
Lattice	Poset where every pair has LUB and GLB	Divisors of n under divisibility	Boolean algebra, circuit design.

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Combinatorics — Recurrences & Generating Functions (CS)

Concept	Formula / Method	When to Use
Linear Recurrence (homogeneous)	$a_n = C_1 a_{n-1} + C_2 a_{n-2}$; char. eqn $r^2 = C_1 r + C_2$; $a_n = A r^n + B s^n$ (distinct roots)	Fibonacci, divide-and-conquer recurrences.
Master Theorem	$T(n) = aT(n/b) + f(n)$; compare $f(n)$ to $n^{\log_b a}$	Algorithm time complexity; D&C recurrences (CS GATE).
Ordinary Generating Function	$G(x) = \sum a_n x^n$; operations on power series	Counting sequences; solving recurrences formally.
Inclusion-Exclusion (general)	$ A_1 \cup \dots \cup A_n = \sum A_i - \sum A_i \cap A_j + \dots$	Counting with overlapping conditions.
Pigeonhole Principle	n items in k boxes $\rightarrow \geq 1$ box has $\geq \lceil n/k \rceil$ items	Existence proofs; graph coloring bounds.
Stirling Numbers (2nd kind)	$S(n, k)$: ways to partition n elements into k non-empty subsets	Set partition counting problems.



Method	Formula / Iteration	Why / Convergence	When to Use
Bisection Method	$x_m = (a+b)/2$; replace a or b based on sign of $f(x_m)$	Guaranteed convergence; slow (linear, halves interval each step).	Finding root when sign change bracketed.
Newton-Raphson Method	$x_{n+1} = x_n - f(x_n)/f'(x_n)$	Quadratic convergence; requires $f'(x)$ and good initial guess.	Fast root finding; most common in GATE.
Regula Falsi	$x = [a \cdot f(b) - b \cdot f(a)] / [f(b) - f(a)]$	Linear interpolation; slower than Newton but more robust.	When f' is hard to compute.
Trapezoidal Rule	$\int_a^b f \, dx \approx h/2[f_0 + 2(f_1 + \dots + f_{n-1}) + f_n]$; $h=(b-a)/n$	Approximates curve with trapezoids; error $O(h^2)$.	Numerical integration with moderate accuracy.
Simpson's 1/3 Rule	$\approx h/3[f_0 + 4f_1 + 2f_2 + 4f_3 + \dots + f_n]$; n must be even	Parabolic fit; error $O(h^4)$ — more accurate than trapezoidal.	Higher accuracy integration; n must be even.
Lagrange Interpolation	$P(x) = \sum y_i \cdot \prod_{j \neq i} (x - x_j) / (x_i - x_j)$	Unique polynomial through n+1 points.	Finding values between known data points.
Euler's Method (ODE)	$y_{n+1} = y_n + h \cdot f(x_n, y_n)$	First-order Taylor approximation; error $O(h)$.	Simple ODE integration; first step in RK methods.
Runge-Kutta (RK4)	$y_{n+1} = y_n + (k_1 + 2k_2 + 2k_3 + k_4)h/6$; $k_1 = f(x_n, y_n)$; $k_2 = f(x_n + h/2, y_n + k_1h/2)$ etc.	4th order accuracy $O(h^4)$; workhorse numerical ODE solver.	Accurate ODE solutions; common in CE/ME GATE.





DA

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CS



Vector Spaces & Subspaces

Concept	Definition / Formula	Why / Meaning	When to Use
Vector Space	Closed under addition & scalar multiplication; 8 axioms	Generalizes $\mathbb{R}^n$; all linear algebra lives here.	Abstract linear spaces; function spaces.
Subspace test	Non-empty; closed under + and scalar $\times$; contains $\mathbf{0}$	Subset that is itself a vector space.	Column space, null space, row space tests.
Linear Independence	$c_1 v_1 + \dots + c_n v_n = \mathbf{0} \rightarrow c_1 = \dots = c_n = 0$	No vector expressible as linear combination of others.	Testing basis; rank determination.
Basis & Dimension	Linearly independent spanning set; $\dim =$ number of basis vectors	Minimal spanning set; unique representation of vectors.	Finding dimension of column/row/null space.
Orthogonality	$u \perp v$ iff $u \cdot v = 0$	Perpendicular vectors; zero inner product.	Projection, least squares, signal decomposition.
Orthonormal Basis	$\langle u_i, u_j \rangle = \delta_{ij}$ (1 if $i=j$, 0 otherwise)	Simplifies all inner product calculations.	Gram-Schmidt process; QR decomposition.
Projection onto subspace	$\text{proj}_W(v) = A(A^T A)^{-1} A^T v$ (A has basis of W as columns)	Nearest point in W to v ; minimizes distance.	Least squares regression; DA dimensionality reduction.
SVD (Singular Value Decomposition)	$A = U \Sigma V^T$; U, V orthogonal; Σ diagonal (singular values)	Generalization of eigendecomposition to non-square matrices.	PCA, dimensionality reduction, matrix approximation (DA).
Projection Matrix	$P = A(A^T A)^{-1} A^T$; $P^2 = P$; $P^T = P$	Idempotent symmetric matrix; projects onto column space of A .	Least squares, regression analysis (DA).

Concept	Definition / Formula	Why / Meaning	When to Use
Quadratic Form	$Q = x^T A x$; PD iff all eigenvalues > 0 ; PSD iff ≥ 0	Classifies geometry of quadric surfaces; optimization.	Optimization, stability analysis, ML loss functions.



Optimization (DA Branch)

Concept	Formula / Method	Why / When
Single Variable Optimization	$f'(x) = 0$ (necessary); $f''(x) > 0 \rightarrow \text{min}$; $f''(x) < 0 \rightarrow \text{max}$	1D loss functions, convex analysis.
Gradient Descent	$x_{n+1} = x_n - \alpha \cdot \nabla f(x_n)$ (α =learning rate)	Iterative minimization in ML/DA; converges for convex f .
Lagrange Multipliers	$\nabla f = \lambda \nabla g$ (constrained optimization: maximize f subject to $g=0$)	Constrained optimization problems.
Convex Function	$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$; Hessian PSD	Local minimum is global minimum; ML loss design.


